Output

--- Step 1: Put-Call Parity and Option Pricing ---

Q1: Put-call parity holds for European options: C + K*exp(-r*T) = P + S0.

LHS (C + K*e^(-rT)) = 103.3718

RHS (P + S0) = 103.3718

Difference = 0.000000

Q2: Solve for C: C = P + S0 - K*exp(-r*T).

Calculated Call Price C: 4.6140

Directly Priced Call C: 4.6140

Difference: 0.000000

Q3: Solve for P: P = C - S0 + K*exp(-r*T).

Calculated Put Price P: 3.3718

Directly Priced Put P: 3.3718

Difference: 0.000000

Q4: For American options strict equality fails due to early exercise; only inequalities hold.

American Options 'Parity' Check:

LHS (C_A + K*e^(-rT)) = 103.3718

RHS (P_A + S0) = 103.4793

Difference = 0.107563

Q5: Price ATM European call and put (binomial)

Steps: 1000

European Call: 4.61

European Put : 3.37

Q6: European Delta at t=0 (binomial)

Call Delta: 0.5694

Put Delta: -0.4306

Q7: Vega (European) via σ bump 20% -> 25%

Call: 4.61 -> 5.60, Vega ≈ 19.6631

Put : 3.37 -> 4.35, Vega ≈ 19.6631

Q8: Price ATM American call and put (binomial)

American Call: 4.61

American Put : 3.48

Q9: American Delta at t=0 (binomial)

Call Delta: 0.5694

Put Delta: -0.4495

Q10: Vega (American) via σ bump 20% -> 25%

Call: 4.61 -> 5.60, Vega ≈ 19.6631

Put : 3.48 -> 4.46, Vega ≈ 19.5986

Q11: European put–call parity check

LHS (C + K e^(-rT)) = 103.3718

RHS (P + S0) = 103.3718

Difference = 0.000000

Q12: American 'parity' equality fails

LHS (C_A + K e^(-rT)) = 103.3718

RHS (P_A + S0) = 103.4793

Difference = 0.107563

Q13: Confirm C_E <= C_A (no dividends) -> typically equal.

Diff (A−E) for Call = 0.000000

Q14: Confirm P_E <= P_A.

Diff (A−E) for Put = 0.107563

--- Step 2: Trinomial Tree and Strike Price Analysis ---

Q15: Price 5 European CALLs using a trinomial tree

Strike	Moneyness	European Call
90	Deep OTM	10.37
95	OTM	6.63
100	ATM	3.81
105	ITM	1.96
110	Deep ITM	0.90

Q16: Price 5 European PUTs using a trinomial tree

Strike	Moneyness	European Put
90	Deep OTM	0.74
95	OTM	1.94
100	ATM	4.06
105	ITM	7.14
110	Deep ITM	11.02

Q17: Price 5 American CALLs using a trinomial tree

Strike	Moneyness	American Call

90	Deep OTM	10.50
95	OTM	6.69
100	ATM	3.84
105	ITM	1.97
110	Deep ITM	0.90

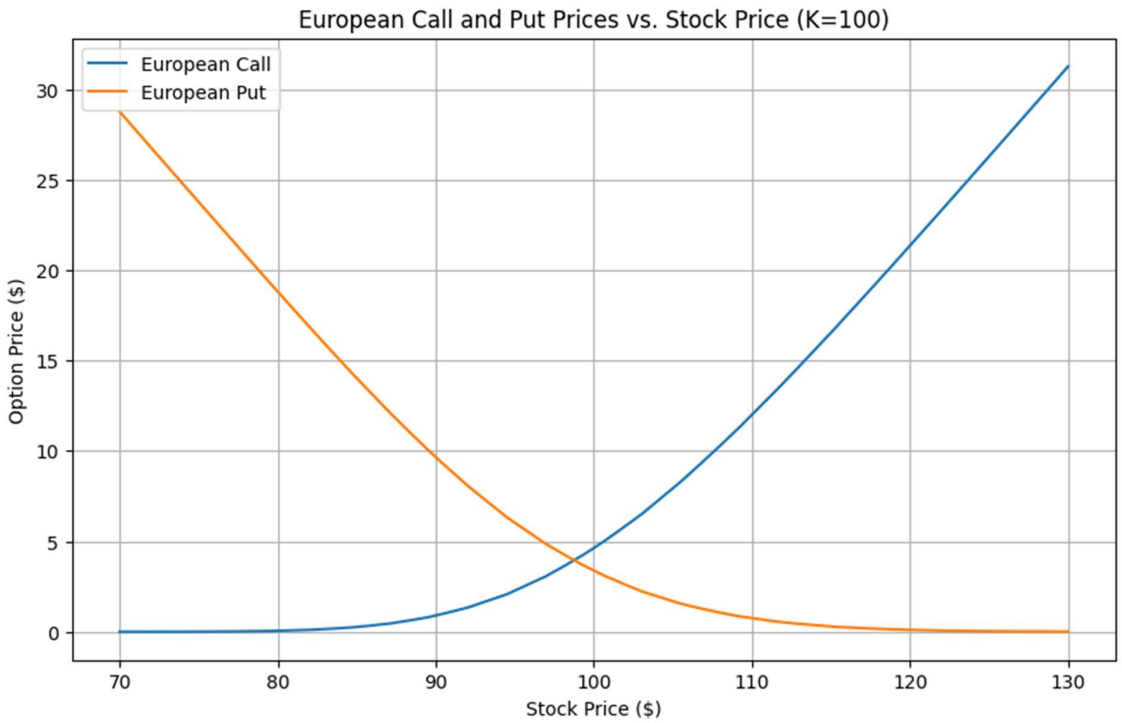
Q18: Price 5 American PUTs using a trinomial tree

Strike	Moneyness	American Put
90	Deep OTM	0.74
95	OTM	1.94
100	ATM	4.06
105	ITM	7.14
110	Deep ITM	11.02

Q19: Graph data — option value vs S0 (European, K=100)

Arrays prepared: val_call_vs_S, val_put_vs_S

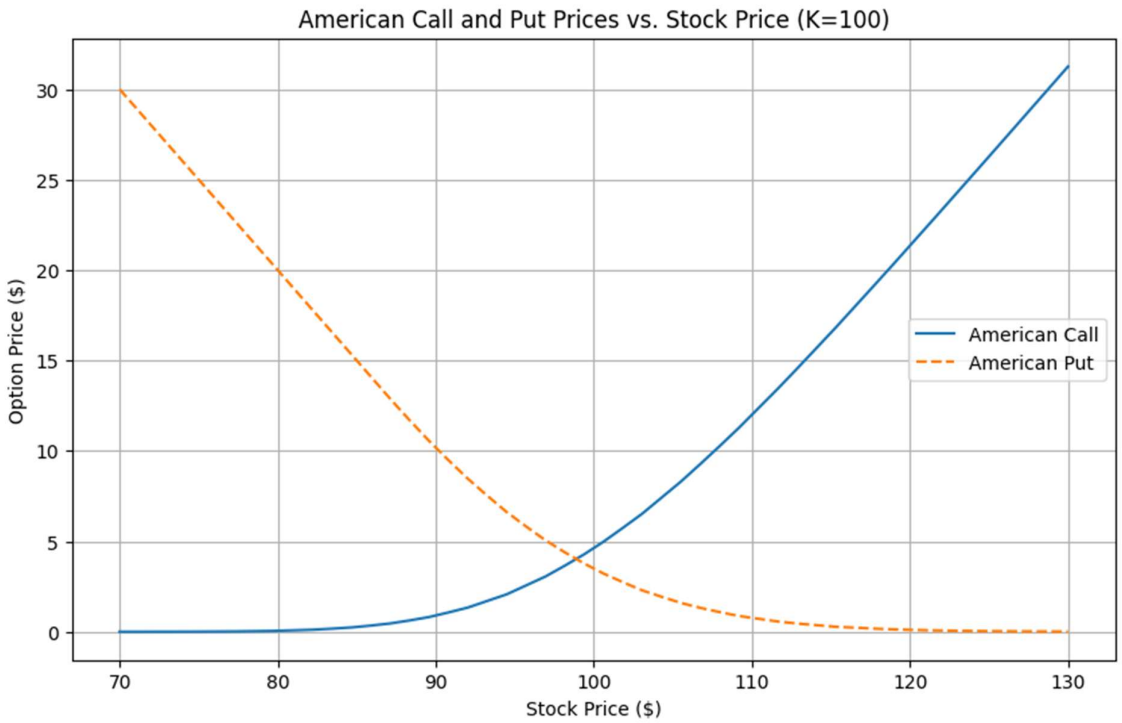
Length: 51 | Sample first 5 call values: [0. 0. 0. 0. 0.]



Q20: Graph data — Delta vs S0 (European, K=100)

Arrays prepared: delta_call_vs_S, delta_put_vs_S

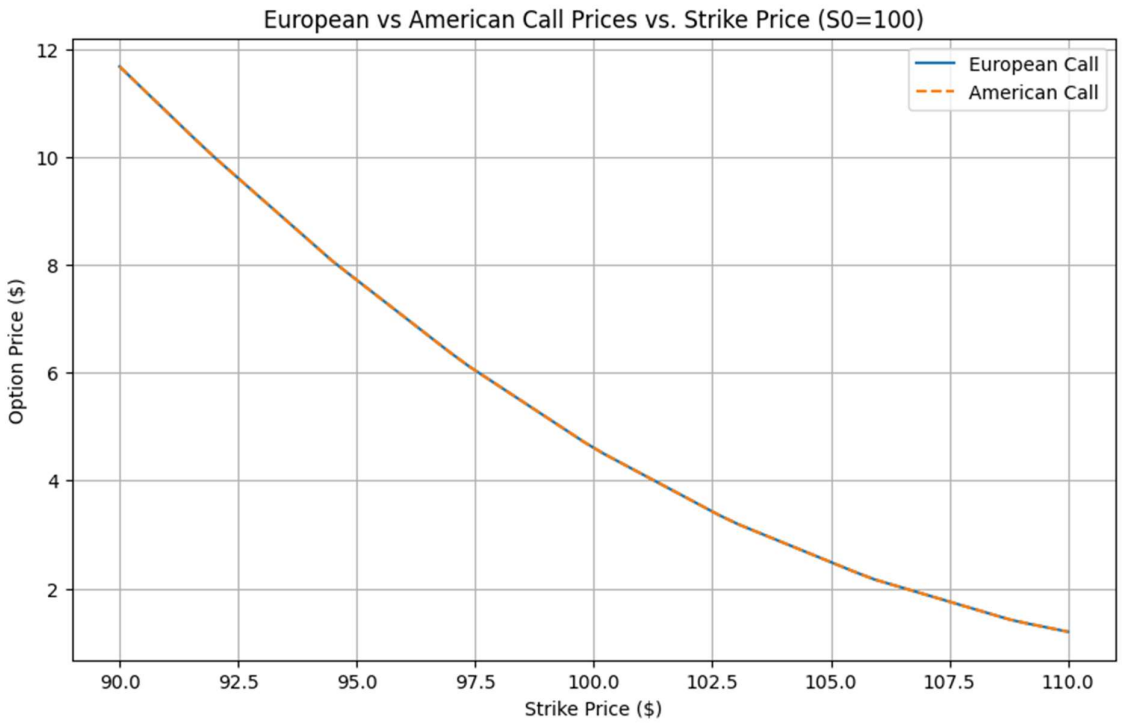
Length: 51 | Sample first 5 call deltas: [0. 0. 0. 0. 0.]



Q21: Graph data — option value vs K (European, S0=100)

Arrays prepared: val_call_vs_K, val_put_vs_K

K values: [90 95 100 105 110] | Call values: [11.6717 7.7176 4.61   2.4819 1.1918]

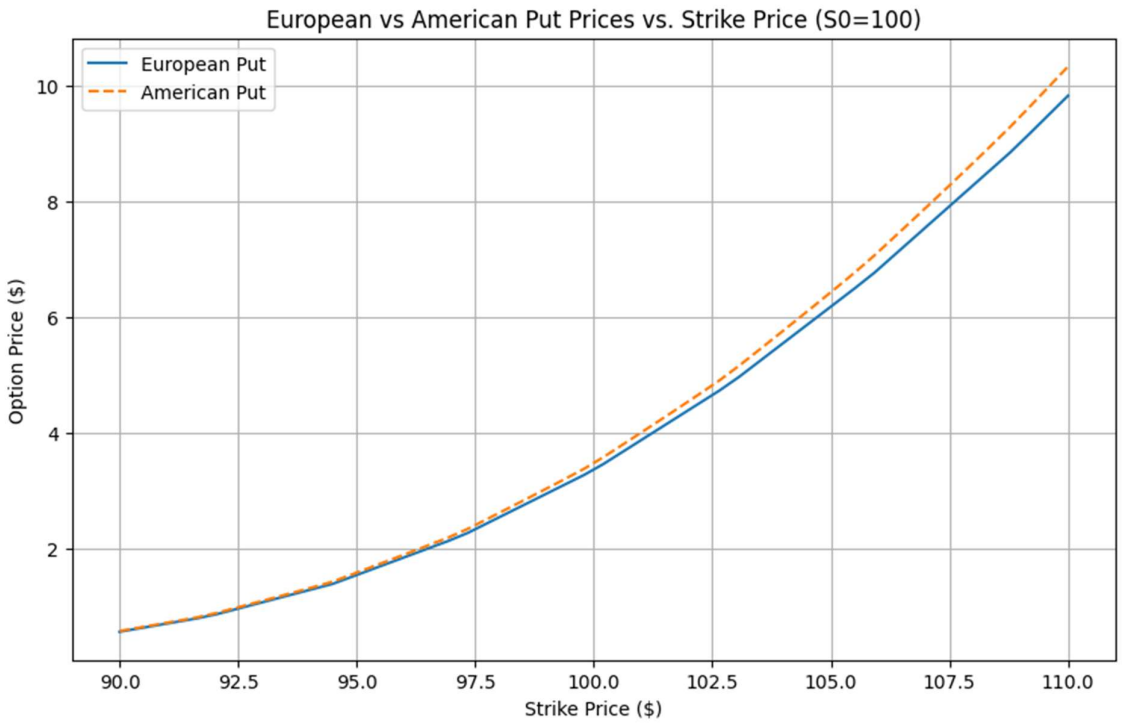


Q22: Graph data — Delta vs K (European, S0=100)

Arrays prepared: delta_call_vs_K, delta_put_vs_K

K values: [90 95 100 105 110] | Call deltas: [0.8903 0.7541 0.5694 0.3771 0.218]

--- Q19-22: Generating Graphs ---



Q23: Check put–call parity for European options (trinomial)

Strike	LHS (C + K e^{-rT})	RHS (P + S0)	|Diff|
90	99.2489	100.7377	1.4888
95	100.4473	101.9361	1.4888
100	102.5680	104.0568	1.4888
105	105.6541	107.1429	1.4888
110	109.5308	111.0196	1.4888

Q24: American 'parity' equality fails (trinomial)

Strike	LHS (C_A + K e^{-rT})	RHS (P_A + S0)	|Diff|
90	99.3812	100.7377	1.3566
95	100.5089	101.9362	1.4272
100	102.5946	104.0572	1.4626
105	105.6646	107.1444	1.4798
110	109.5346	111.0244	1.4898

--- Step 3: Real-World Questions ---

Q25: 3-step dynamic Delta hedging (European Put, binomial)

Stock tree (rows=#downs, cols=time):

```
[[180.        199.34139817 220.76107236 244.48233791]
 [  0.        162.53523    180.        199.34139817]
 [  0.          0.        146.76500551 162.53523   ]
 [  0.          0.          0.        132.5249107 ]]
```

European put values (same indexing):

```
[[13.8223851   5.01268419  0.         0.       ]
 [ 0.        22.4055917   9.87779048  0.       ]
 [ 0.         0.        34.62933781 19.46477   ]
 [ 0.         0.         0.        49.4750893 ]]
```

Delta at each node (same indexing, last col empty):

```
[[-0.47255415 -0.24233392  0.       ]
 [ 0.        -0.74474354 -0.52884533]
 [ 0.         0.        -1.       ]]
```

Q26: 25-step American Put — price and pathwise Δ idea

American Put Price (25-step binomial): 13.04

For hedging, compute Δ at each node similarly to Q25, but with early exercise check.

Q27: Asian ATM Put via Monte Carlo (arithmetic average)

Asian ATM Put Price (MC): 7.75

(Averaging reduces effective volatility -> typically cheaper; smoother hedging.)

Final Summary: European vs. American (Trinomial, all strikes)

Strike	Eur Call	Am Call	Eur Put	Am Put
90	10.37	10.50	0.74	0.74
95	6.63	6.69	1.94	1.94
100	3.81	3.84	4.06	4.06
105	1.96	1.97	7.14	7.14
110	0.90	0.90	11.02	11.02